Evaluacion

May 25, 2026

[2]:
```
# 1. Importar librerías
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score,␣
 ↪f1_score, classification_report

# 2. Cargar el dataset
df = pd.read_csv('trafico_red.csv')

# 3. Eliminar columnas no numéricas (IPs, timestamp, etc.)
columnas_a_eliminar = ['src_ip', 'dst_ip', 'timestamp']
for col in columnas_a_eliminar:
    if col in df.columns:
        df = df.drop(col, axis=1)

print("Dataset cargado. Primeras filas:")
print(df.head())
print("\nDistribución de etiquetas:")
print(df['label'].value_counts())

# 4. Preparar los datos
X = df.drop('label', axis=1)
y = df['label']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,␣
 ↪random_state=42)
print(f"\nEntrenamiento: {X_train.shape[0]} muestras")
print(f"Prueba: {X_test.shape[0]} muestras")

# 5. Entrenar y evaluar modelos
modelos = {
    "Random Forest": RandomForestClassifier(n_estimators=100, random_state=42),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "KNN": KNeighborsClassifier(n_neighbors=5)
}
```

```python
for nombre, modelo in modelos.items():
    modelo.fit(X_train, y_train)
    y_pred = modelo.predict(X_test)

    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred, average='weighted')
    recall = recall_score(y_test, y_pred, average='weighted')
    f1 = f1_score(y_test, y_pred, average='weighted')

    print(f"\n{'='*40}")
    print(f"--- {nombre} ---")
    print(f"Accuracy:  {accuracy:.4f}")
    print(f"Precision: {precision:.4f}")
    print(f"Recall:    {recall:.4f}")
    print(f"F1-Score:  {f1:.4f}")
    print("\nReporte de Clasificación:")
    print(classification_report(y_test, y_pred))
```

```
Dataset cargado. Primeras filas:
   src_port  dst_port  protocol  flow_duration   flow_byts_s  flow_pkts_s  \
0     51913       443        17      25.285889     35.869809     0.395477
1      5353      5353        17      19.989860     15.457837     0.150076
2     29097        53        17       0.020400  23333.333333   147.058824
3     40920       443         6       5.554403   2636.286924     7.921643
4     49336       443         6      10.577655    177.449539     0.661772

   fwd_pkts_s  bwd_pkts_s  tot_fwd_pkts  tot_bwd_pkts  …  fwd_blk_rate_avg  \
0    0.237286    0.158191             6             4  …               0.0
1    0.150076    0.000000             3             0  …               0.0
2   98.039216   49.019608             2             1  …               0.0
3    4.140859    3.780784            23            21  …               0.0
4    0.661772    0.000000             7             0  …               0.0

   bwd_blk_rate_avg  fwd_seg_size_avg  bwd_seg_size_avg  cwr_flag_count  \
0      0.000000e+00         94.000000         85.750000               0
1      0.000000e+00        103.000000          0.000000               0
2      0.000000e+00         80.000000        316.000000               0
3      1.636425e+07        221.130435        455.095238               0
4      0.000000e+00        268.142857          0.000000               0

   subflow_fwd_pkts  subflow_bwd_pkts  subflow_fwd_byts  subflow_bwd_byts  \
0                 6                 4               564               343
1                 3                 0               309                 0
2                 2                 1               160               316
3                23                21              5086              9557
4                 7                 0              1877                 0
```

```
    label
0  normal
1  normal
2  normal
3  normal
4  normal

[5 rows x 80 columns]

Distribución de etiquetas:
label
normal     1681
psiphon     164
Name: count, dtype: int64

Entrenamiento: 1291 muestras
Prueba: 554 muestras

========================================
--- Random Forest ---
Accuracy:  0.9440
Precision: 0.9398
Recall:    0.9440
F1-Score:  0.9394

Reporte de Clasificación:
              precision    recall  f1-score   support

      normal       0.95      0.99      0.97       499
     psiphon       0.82      0.56      0.67        55

    accuracy                           0.94       554
   macro avg       0.88      0.77      0.82       554
weighted avg       0.94      0.94      0.94       554


========================================
--- Decision Tree ---
Accuracy:  0.9224
Precision: 0.9256
Recall:    0.9224
F1-Score:  0.9239

Reporte de Clasificación:
              precision    recall  f1-score   support

      normal       0.96      0.95      0.96       499
     psiphon       0.60      0.65      0.63        55
```

```
    accuracy                           0.92       554
   macro avg       0.78      0.80      0.79       554
weighted avg       0.93      0.92      0.92       554


========================================
--- KNN ---
Accuracy:  0.9170
Precision: 0.9040
Recall:    0.9170
F1-Score:  0.9052

Reporte de Clasificación:
              precision    recall  f1-score   support

      normal       0.93      0.98      0.96       499
     psiphon       0.66      0.35      0.45        55

    accuracy                           0.92       554
   macro avg       0.79      0.66      0.70       554
weighted avg       0.90      0.92      0.91       554
```

[ ]: